

Useful Lists of Numbers

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1 Introduction

This document will summarise all the important sequences that you will probably need. Please make an effort to remember them, as these will come up a lot a lot during your time in intermediate.

2 Sequences

Theorem 2.1 – Prime Numbers

2, 3, 5, 7, 11, 13, 17, 19, 23, 29, 31, 37, 41, 43, 47, 53, 59, 61, 67, 71, 73, 79, 83, 89, 97, ...

These are all the prime numbers less than 100. A few useful facts:

- 1: Prime numbers are numbers that have exactly 2 numbers that divide it: 1 and itself.
- 2: There are exactly **25** prime numbers less than 100.
- 3: Some numbers like 51, 57, 87, 91 may "look" like prime numbers, but they are not.

Theorem 2.2 – Square Numbers

1, 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144, 169, 196, ...

It's useful to remember these up to the number 12^2 .

Theorem 2.3 – Cube Numbers

1, 8, 27, 64, 125, 216, 343, 512, 729, 1000, 1331, 1728, ...

It's best if you can remember them up to $10^3 = 1000$.

Theorem 2.4 – Triangular Numbers

1, 3, 6, 10, 15, 21, 28, 36, 45, 55, 66, 78, ...

One of the most important sequences.

See the next page to see how this sequence is constructed.

$$1 = 1$$

$$3 = 1 + 2$$

$$6 = 1 + 2 + 3$$

$$10 = 1 + 2 + 3 + 4$$

$$15 = 1 + 2 + 3 + 4 + 5$$

and so on.

If you imagine triangles made out of dots, the triangular numbers are the number of dots you will need for each triangle.

Theorem 2.5 – Fibonacci Sequence

$$1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, \dots$$

The sequence starts with 1 and 1. Then, each next number is equal to the previous 2 numbers added together. This actually leads to a lot of very cool things later on, but for now, just remember that a sequence like this exists and is very famous. Make sure you can recognise it, if you ever comes up in a problem.